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ORIGINAL RESEARCH

A Gaussian Unscented Kalman Filter algorithm for indoor positioning system using Ultra Wide Band measurement

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Abstract

In order to further improve the accuracy of the non-linear positioning model in the research of ultra wide band (UWB) indoor positioning, a Gaussian unscented Kalman filter (GUKF) algorithm is proposed in this paper. This localisation algorithm first uses a Gaussian function to design a Gaussian smoothing filter template to process the smoothing of experimental data in the GUKF algorithm, and then the filtering algorithm is used to obtain higher positioning accuracy. This paper utilises simulations and actual experiments to verify and analyse the GUKF algorithm, and the actual experiment environment was divided into line-of-sight (LOS) and non-line-of-sight (NLOS) experimental environments. The measured experimental results indicate that in the static test of location tags in LOS and NLOS experimental environments, the root mean square error (RMSE) of the GUKF algorithm is reduced by 15.88% and 14.10%, respectively; in the dynamic test, the RMSE of the GUKF algorithm is reduced by 16.67% and 17.89%, respectively, compared with the unscented Kalman filter algorithm. In addition, the positioning performance evaluation method of the mean error and cumulative distribution function curve also verifies that the GUKF algorithm has a higher positioning accuracy than the UKF, Least Squares, and Time of Arrival algorithms.

KEYWORDS

Gaussian distribution, Kalman filters, position measurement, smoothing methods, ultra wideband technology

1 | INTRODUCTION

With rapid advancements in science and technology, the demand for location-based services is increasing. Positioning and navigation technology is now widely used in areas such as factory personnel management, unmanned driving, emergency rescue, and urban intelligent transportation. Outdoor positioning often relies on satellite systems such as GPS [1–3], Galileo, and BeiDou [4] due to their real-time, wide-range, and accurate all-weather capabilities. However, as people spend most of their time indoors, research into high-precision indoor positioning technology is becoming crucial. This technology is essential for managing movement and specialised staff in crowded environments such as airports, hospitals, and nursing homes, and for providing enhanced navigation services through precise positioning data.

While the field of global satellite positioning technology is relatively developed, signal reception strength is affected by building obstructions. Additionally, receiving a satellite positioning signal in an indoor environment, like a basement, can be challenging, leading to reduced or non-existent positioning accuracy. Therefore, satellite positioning technology cannot meet the growing needs of indoor positioning technology [5–8].

The main contributions of this paper are summarised as follows.

1. *Introducing Gaussian filtering optimised unscented Kalman filter (UKF) algorithm*: The use of Gaussian traceless

Kalman filter (KF) algorithm is proposed to improve the accuracy of indoor positioning systems. By comparing with the traditional KF and other advanced filtering techniques, it is demonstrated that the GUKF algorithm has significant advantages in improving the positioning accuracy
2. *Hardware design combined with algorithm*: The double-sided two-way ranging (DS-TWR) ranging model is used in a self-designed hardware device to improve the ranging accuracy. This demonstrates the effective integration of the algorithm with the actual hardware, which provides technical support for achieving high-precision indoor positioning.
3. *Experimental validation*: The proposed GUKF positioning algorithm is simulated and validated through dynamic and static experiments, and is tested practically in Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) environments using self-designed hardware devices, respectively. The experimental results show that the algorithm can maintain high positioning accuracy under different environmental conditions.

The thesis is arranged as follows: Section 2 introduces the positioning principle of ultra wide band (UWB), in which the ranging model is compared and analysed, and the DS-TWR ranging model is used in the hardware equipment designed by ourselves to improve the ranging accuracy. Section 3 introduces the establishment of the non-linear model of UWB localisation algorithm and the derivation of GUKF algorithm are described. In Section 4, the dynamic and static experiments of the proposed GUKF positioning algorithm are simulated and verified, and the actual experiments are analysed and verified by self-designed hardware equipment, in which the actual experiments are conducted in LOS and NLOS environments, respectively. Section 5 is the conclusion.

2 | RELATED WORK

Currently, the most commonly used positioning technologies in indoor positioning systems are primarily WiFi positioning [9, 10], wireless Bluetooth positioning, and ZigBee [11]. While these wireless technologies can satisfy certain indoor positioning requirements, they often face significant challenges such as large positioning errors, poor signal penetration, and inadequate transmission distances [12, 13]. In contrast, UWB positioning technology offers distinct advantages, including high positioning accuracy, low implementation complexity, and robust signal anti-interference capabilities [14–17]. UWB utilises narrow pulse waves at the nanosecond level for ultra-wideband transmission, making it particularly suitable for high-demand indoor positioning applications [18, 19]. Recent research studies by various researchers have focused on UWB positioning technology, covering aspects such as positioning algorithms and products. These efforts have yielded promising results, highlighting UWB's potential to address the limitations of existing indoor positioning technologies effectively.

Generally, the position coordinate solution model established in the measurement system of UWB original data is a non-linear model, so non-linear filtering algorithms are often used to optimise the location information of the positioning tag. In the literature, a positioning system composed of UWB positioning technology, inertial measurement unit (IMU), and distance sensors is proposed [20]. The sensor information of various positioning devices is multi-fused, and the position information is calculated using the extended Kalman filter (EKF) algorithm. Experimental results show a maximum plane positioning error of 20 cm. Ref. [21] proposes a new method combining the KF algorithm and arrival time-based location algorithm, demonstrating a 37% reduction in root mean square error (RMSE) compared to traditional methods. Ref. [22] presents an EKF ultra-wideband indoor positioning method, achieving sub-metre accuracy in complex indoor environments. The main idea is to use the KF algorithm to calculate the non-linear model using the Taylor expansion, eliminating terms above the second order. Ref. [23] suggests a positioning scheme that integrates UWB and IMU, using an improved distributed iterative EKF algorithm for fusion information processing. Experimental results indicate that the improved EKF algorithm has higher positioning accuracy than the traditional EKF algorithm. Although the EKF algorithm can achieve good positioning results, the calculation of the Jacobian matrix is complicated. The UKF based on the unscented transformation (UT) transform, proposed in ref. [24], uses the UT to approximate the posterior probability density function of the non-linear system, reflecting system characteristics more accurately without the need to linearise the model.

Based on this, further research studies have focused on using deep learning and more accurate positioning algorithms to improve the performance of UWB indoor positioning. Literature [25] proposes a UWB indoor positioning method based on deep learning long short-term memory (LSTM) networks. The method exploits the penetration property of UWB signals to reduce the impact of multipath effect on positioning accuracy. By predicting the user's location through a LSTM network, an average positioning error of 7 cm is achieved, which is a significant improvement over traditional UWB positioning methods.

In addition, the literature [26] analysed the positioning error of indoor positioning systems based on UWB measurements. The study reviewed different positioning algorithms and compared their performance experimentally. The results show that the fingerprint estimation algorithm performs best in terms of positioning error, while the linearised least squares (LS) algorithm performs poorly.

Finally, literature [27] proposes an accurate indoor user location estimator for multi-reference point UWB localisation. The method employs the EKF to effectively deal with challenges such as multipath effects and clock drift. Experimental results show that the method proves its effectiveness by minimising the positioning error in multi-anchor point UWB indoor positioning.

In order to further improve the positioning accuracy of UWB non-linear model, a GUKF localisation algorithm based on Gaussian smoothing filter is proposed. Gaussian function is used to generate a Gaussian template to weight the data collected at each moment, and then UKF filtering is performed on the processed experimental data to improve the positioning accuracy.

3 | POSITIONING ALGORITHM

3.1 | UWB positioning principle

UWB indoor localisation uses a TOA scheme. The distance d between the tag and the base station can be calculated based on the delay estimation. Assuming there is no range error, three circles intersect at a point as shown in Figure 1. In the figure, the base station is denoted as BS_i corresponding to the co-ordinates (x_i, y_i). The localisation tag Tag corresponds to coordinates (x, y), and the distance between it and the base station BS_i is d_i $(i = 1, 2, 3, 4)$.

In practice, there will be hardware errors that cause the three circles to intersect in an area rather than a point. Combining the TOA localisation methods results in Equations (1–3).

$$\sqrt{(x_1 - x)^2 - (y_1 - y)^2} = d_1 \tag{1}$$

$$\sqrt{(x_2 - x)^2 - (y_2 - y)^2} = d_2 \tag{2}$$

$$\sqrt{(x_3 - x)^2 - (y_3 - y)^2} = d_3 \tag{3}$$

Equations (1) and (2) are squared and differenced to give Equation (4),

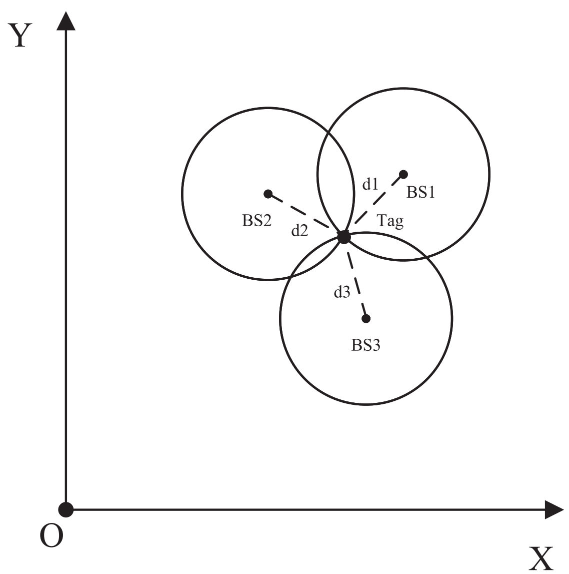

FIGURE 1 Time of Arrival geometric positioning model.

$$\begin{aligned} d_2^2 - d_1^2 = {} & \left(x_2^2 + y_2^2\right) - \left(x_1^2 + y_1^2\right) \\ & -2(x_2 - x_1)\,x - 2\left(y_2 - y_1\right) y \end{aligned} \tag{4}$$

and similarly Equation (5).

$$\begin{aligned} d_3^2 - d_1^2 = {} & \left(x_3^2 + y_3^2\right) - \left(x_1^2 + y_1^2\right) \\ & -2(x_3 - x_1)\,x - 2\left(y_3 - y_1\right) y \end{aligned} \tag{5}$$

Combining Equations (4) and (5) yields the matrix Equation (6),

$$\begin{aligned} & \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \\ & \frac{1}{2} \begin{bmatrix} \left(x_2^2 + y_2^2\right) - \left(x_2^2 + y_2^2\right) + d_1^2 - d_2^2 \\ \left(x_3^2 + y_3^2\right) - \left(x_2^2 + y_2^2\right) + d_1^2 - d_3^2 \end{bmatrix} \end{aligned} \tag{6}$$

which in turn yields Equation (7).

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} = {} & \frac{1}{2} \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix}^{-1} \\ & \begin{bmatrix} \left(x_2^2 + y_2^2\right) - \left(x_2^2 + y_2^2\right) + d_1^2 - d_2^2 \\ \left(x_3^2 + y_3^2\right) - \left(x_2^2 + y_2^2\right) + d_1^2 - d_3^2 \end{bmatrix} \end{aligned} \tag{7}$$

And if $i \geq 4$, that is, there are more than three base stations BS_i, the problem can be solved using an algorithm such as the LS algorithm. For the measurement in 4 BS cases, a Gaussian smoothing filter-based UKF (GUKF) algorithm is proposed in this section to cope with the data drift caused by large noise signals and complex environmental factors during ranging.

3.2 | Establishment of UWB positioning model

Within the UWB positioning area defined by the base stations, a state equation model is established based on the position, velocity and acceleration information of the positioning tag over discrete time intervals. This model is given by the following equation:

$$X_k = AX_{k-1} + W \tag{8}$$

In Equation (8), X is the state vector, A is the state transition matrix, and W is the noise matrix. The state vector X is expressed as follows:

$$X = \begin{bmatrix} P_x & P_y & V_x & V_y \end{bmatrix} \tag{9}$$

In Equation (9), P_x and V_x denote the position and velocity information of the tag in the x-axis direction at the K moment, and P_y and V_y denote the position and velocity information of the tag in the y-axis direction a t the K moment.

In Equation (8), where the state transition matrix A is

$$A = \begin{bmatrix} 1 & 0 & t & 0 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

where t is the interval of data collection.

In this positioning experiment, a total of four BS constitute the UWB positioning area. The ranging information between the four base stations and the tag can be calculated according to the estimated tag position and known coordinate information of the base station at the previous time Equation (11):

$$R_{i,k} = \sqrt{(x_k - x_i) + (y_k - y_i)} + n_{i,k} \tag{11}$$

where $R_{i,k}$ is the distance between BS_i and the tag at time k, and (x_k, y_k) is the estimated planar coordinate value at time k. (x_i, y_i) is the actual coordinates of BS_i and $n_{i,k}$ is the measurement noise, so the measurement equation of the localisation model is expressed as the following equation:

$$z_k = g(X_k, u_k) = HX_k + u_k \tag{12}$$

In Equation (12), z_k is the measured value at time k, corresponding to Equation (13). H is the measurement transfer matrix. Corresponds to Equation (14). u_k is the measurement noise at time k.

$$z_k = [R_{1,k}, R_{2,k}, R_{3,k}, R_{4,k}] \tag{13}$$

$$H = \begin{bmatrix} \sqrt{(x_k - x_1) + (y_k - y_1)} \\ \sqrt{(x_k - x_2) + (y_k - y_2)} \\ \sqrt{(x_k - x_3) + (y_k - y_3)} \\ \sqrt{(x_k - x_4) + (y_k - y_4)} \end{bmatrix} \tag{14}$$

3.3 | Design of GUKF positioning algorithm

In the actual ranging process, the data of the positioning system is updated quickly, and the data of the neighbouring moments are approximately normally distributed. A Gaussian template function $N(0,\sigma)$ of length $2n+1$ is used to calculate the distance information by circular convolution to update the experimental data.

The Gaussian template function conforming to $N(0,\sigma)$ is the Equation (15):

$$h(x) = \frac{1}{2\pi\sigma^2} e^{\frac{-r^2}{2\sigma^2}} \tag{15}$$

where r^2 represents the distance between other points in the neighbourhood and the centre point, and σ is the standard deviation of the Gaussian function. When the standard deviation is smaller, the Gaussian filtering effect is less obvious, and when the standard deviation is larger, the filtering effect is more obvious.

When $n = 2$ the length of the Gaussian template is 5, then the operator generated by the Gaussian function is Equation (16):

$$h = [h(-2) \quad h(-1) \quad h(0) \quad h(1) \quad h(2)] \tag{16}$$

Set the distance between the localised tag and the four BS as $R_{i,k}$ with $i = 1,2,3,4$ at the moment k. The Gaussian processed distance information $R'_{i,k}$ is shown in Equation (17), and then replace $R_{i,k}$ in Equation (13)

$$R'_{i,k} = \begin{bmatrix} R_{1,k-2} & R_{1,k-1} & R_{1,k} & R_{1,k+1} & R_{1,k+2} \\ R_{2,k-2} & R_{2,k-1} & R_{2,k} & R_{2,k+1} & R_{2,k+2} \\ R_{3,k-2} & R_{3,k-1} & R_{3,k} & R_{3,k+1} & R_{3,k+2} \\ R_{4,k-2} & R_{4,k-1} & R_{4,k} & R_{4,k+1} & R_{4,k+2} \end{bmatrix} \begin{bmatrix} h(-2) \\ h(-1) \\ h(0) \\ h(1) \\ h(2) \end{bmatrix} \tag{17}$$

In Equation (17), $k \in \mathbb{Z}^+ \cap (2, +\infty)$.

The UT of the non-linear system state quantities is required as follows:

$$\overline{X^0} = \hat{x}_0 \tag{18}$$

$$X^i = \overline{X^0} + [\sqrt{P(n+\lambda)}]_i, (i = 1,2\ldots,n) \tag{19}$$

$$X^i = \overline{X^0} - [\sqrt{P(n+\lambda)}]_i, (i = n+1,\ldots 2n) \tag{20}$$

In Equations (18–20), P and $\hat{x}_0$ are the covariance and mean of the state quantities. X^i is the Sigma point set, $\lambda = \alpha^2(n+\kappa) - n$ is the is the scaling factor. Then, the weights of the mean value and covariance of each sampling point in the point set are calculated, as shown in Equations (21–23):

$$W_m^0 = \frac{\lambda}{n+\lambda} \tag{21}$$

$$W_c^0 = \frac{\lambda}{n+\lambda} + \left(1 - \alpha^2 + \beta\right) \tag{22}$$

$$W_m^i = W_c^i = \frac{1}{2(n+\lambda)}, (i = 1,2,\cdots n) \tag{23}$$

In Equations (24) and (25), W_m is the weight of the sampling point, W_c is the weight of the sampling point covariance, and β is set to 2. α is a small positive number in the range of $10^{-4} \le \alpha \le 1$ and the parameter κ. The UKF calculation process is as follows:

Step 1: Set the filter initial value.

$$\hat{X}^0 = E\left(X^0\right) \tag{24}$$

$$P_0 = E\left[\left(X - X_0\right)\left(X - X_0\right)^T\right] \tag{25}$$

where $\overline{X^0}$ is the state quantity and P is the covariance matrix.

Step 2: Time update. Calculate the predicted state quantity and covariance matrix at time k.

$$X_{k+1|k}^i = f\left(X_k^i, u_k\right) \tag{26}$$

$$\hat{X}_{k+1|k}^i = \sum_{i=0}^{2n} W_m^i X_{k+1|k}^i \tag{27}$$

$$\begin{aligned} P_{k|k+1} = \sum_{i=0}^{2n} W_m^i \left(X_{k|k+1}^i - \hat{X}_{k+1|k}^i\right) \\ -\left(X_{k|k+1}^i - \hat{X}_{k+1|k}^i\right)^T + Q \end{aligned} \tag{28}$$

Step 3: Measurement update. Obtain the state vector and covariance matrix at time k.

$$z_{k+1}^i = g\left(X_{k+1}^i, u_{k+1}\right) \tag{29}$$

$$\hat{z}_{k+1}^i = \sum_{i=0}^{2n} W_m^i z_{k+1}^i \tag{30}$$

$$\begin{aligned} P_{k+1}^{zz} = \sum_{i=0}^{2n} W_c^i \left(z_{k+1}^i - \hat{z}_{k+1}^i\right) \\ \cdot \left(z_{k+1}^i - \hat{z}_{k+1}^i\right)^T + R \end{aligned} \tag{31}$$

$$\begin{aligned} P_{k+1}^{xz} = \sum_{i=0}^{2n} W_c^i \left(X_{k+1}^i - \hat{X}_{k+1}^i\right) \\ \cdot \left(z_{k+1}^i - \hat{z}_{k+1}^i\right)^T \end{aligned} \tag{32}$$

where P_{k+1}^{zz} is the prediction covariance matrix and P_{k+1}^{xz} is the interaction covariance matrix. R is the measurement noise, its value is too large or too small will cause the filter divergence, so it is necessary to set a suitable value. This algorithm sets the value to $0.01I$, where I is the identity matrix of 4×4.

Step 4: Calculate the gain factor K and update the state quantity and error covariance.

$$K_{k+1} = P_{k+1}^{xz}\left(P_{k+1}^{zz}\right)^{-1} \tag{33}$$

$$\hat{X}_{k+1} = \hat{X}_{k|k+1} + K_{k+1}\left(z_{k+1} - \hat{z}_{k+1}\right) \tag{34}$$

In Equations (33) and (34), K_{k+1} is the Kalman gain coefficient, $\hat{X}_{k+1}$ is the state quantity at time $k+1$, and $\hat{z}_{k+1}$ is the predicted measurement information.

Step 5: Update the covariance matrix.

$$P_{k+1} = P_{k+1|k} - K_{k+1} P_{k+1}^{zz} K_{k+1}^T \tag{35}$$

where P_{k+1} is the covariance matrix at time $k+1$. As illustrated in Figure 2, the aforementioned algorithmic procedure is represented as a flowchart. Iterative recursion of the above steps can obtain the optimal location estimate of the location tag at any time.

4 | EXPERIMENTAL VERIFICATION AND ANALYSIS

In order to better illustrate the effectiveness of the GUKF filtering localisation algorithm proposed in this paper. Experimental verification and analysis will be carried out, and the experimental part consists of simulation experiments and field experiments. In order to verify that the proposed method has a better positioning effect, field experiments will be conducted in LOS and NLOS environments respectively. For each experiment, the static and dynamic experimental tests of positioning tags were designed. TOA, least-square algorithm (LS) adaptive robust Kalman algorithm (ARKF) and UKF algorithm are introduced to test the performance of positioning algorithm. The performance of the localisation algorithm is evaluated by the RMSE, the mean error and the cumulative distribution function (CDF) of the error.

When comparing the GUKF algorithm with other algorithms, the CDF curve can be used to visualise their performance. The CDF curve shows the probability distribution of a variable over a given range. Its horizontal coordinate indicates the prediction error and the vertical coordinate is the cumulative probability. For a high accuracy algorithm, the CDF curve will be very steep in the range where the prediction error is small, that is, the cumulative probability increases rapidly in the smaller error range. For a less accurate algorithm, the curve will be flatter and the cumulative probability will increase more slowly.

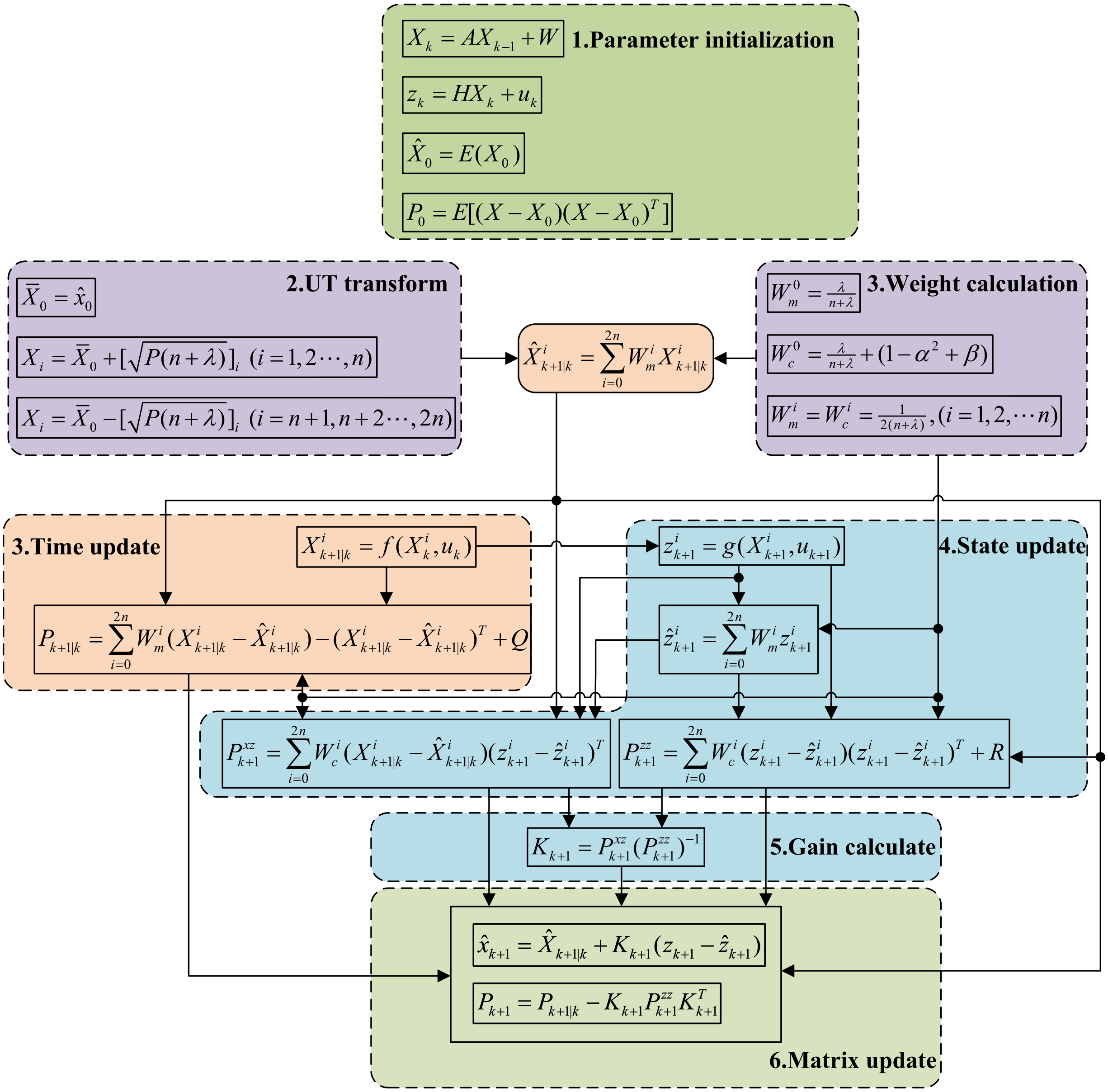

FIGURE 2 Flowchart of the algorithm.

4.1 | Simulation experiment

In generating the simulated UWB datasets, we used standard noise generation methods and localisation simulations, which are widely used in research in related fields [28], although the specific simulation details are not directly based on the IEEE 802.15.4a standard. In the simulation experiment environment, 4 base stations and 1 positioning tag are arranged. The coordinates of base stations are $BS1(0,0)$, $BS2(900,0)$, $BS3(900,1000)$, and $BS4(0,950)$, and the unit is cm. It is known that the range accuracy of the simulated UWB positioning system is $\pm 15cm$, which further makes the randomly generated range error conform to the Gaussian distribution of $N(0,15)$. The length of the Gaussian template used in the smoothing and filtering of experimental data is 5, $\sigma = 1$. In this simulation experiment environment, static experiment and dynamic experiment of positioning tags were carried out, respectively, and the data acquisition frequency of the simulation experiment was $1Hz$.

As shown in Figure 3, the static real data of the location tag is collected in the preset environment for a period of time, using the UKF algorithm, the LS algorithm, the TOA algorithm and the ARKF algorithm, respectively. The GUKF algorithm mentioned in this paper are solved. The red dot in the figure is the real position of the tag, the real point coordinate is $T(450,450)$. Among them, the GUKF prediction points are the most concentrated and close to the real points, followed by the ARKF and UKF prediction points, while the LS and TOA prediction points are still near the real points on the whole, but they are too dispersed and less effective.

Figure 4 is the Euclidean distance error of the static experiment corresponding to the change of the tag over time, between 0 and 100s, GUKF always has an error of less than 5 cm; whereas UKF has a larger error at the beginning, and it also reduces the error around 30s; ARKF maintains an error of less than 10 cm, and as for the errors of LS and TOA they fluctuate greatly between 10 and 50 cm.

Table 1 shows the mean error and RMSE of the different localisation algorithms in the static simulation experiments. It can be seen that the GUKF algorithm performs the best in terms of mean error (1.6303 cm) and root mean squared error (1.8438 cm) and it is significantly lower than the other algorithms. The mean error for the UKF algorithm is 3.2797 cm and the RMSE is 4.5574 cm; although it also performs well, it is still significantly lower than the GUKF. RMSE is 4.5574 cm, which also performs well but still has a significant gap compared to GUKF. LS algorithm has a mean error of 14.0573 cm and RMSE of 16.0183 cm, which indicates that it performs poorly in dealing with non-linear problems. TOA algorithm has a mean error of 18.4620 cm and RMSE of 20.7913 cm, which is the highest among all algorithms, indicating that it performs poorly in dealing with complex problems. The ARKF algorithm has a mean error of 3.8268 cm and an RMSE of 6.4628 cm, which is slightly higher than that of the GUKF, but still remains at a low level, showing its advantage in robustness. Overall, the GUKF algorithm shows the highest positioning accuracy and stability in the static simulation, and is suitable for scenarios that require high-precision positioning.

Figure 5 shows the CDF curves drawn according to the errors of different algorithms in static simulation experiments. It can be seen from the figure that the CDF curve of the GUKF algorithm proposed in this paper is in the upper left of the other four algorithms, indicating that the coordinate error value solved by the GUKF algorithm has a higher probability of being lower than the threshold value when the same error threshold is set. In this graph, GUKF's CDF curve is steeper, indicating that the coordinates calculated using this algorithm are denser and the error variation is minimal. Therefore, in the

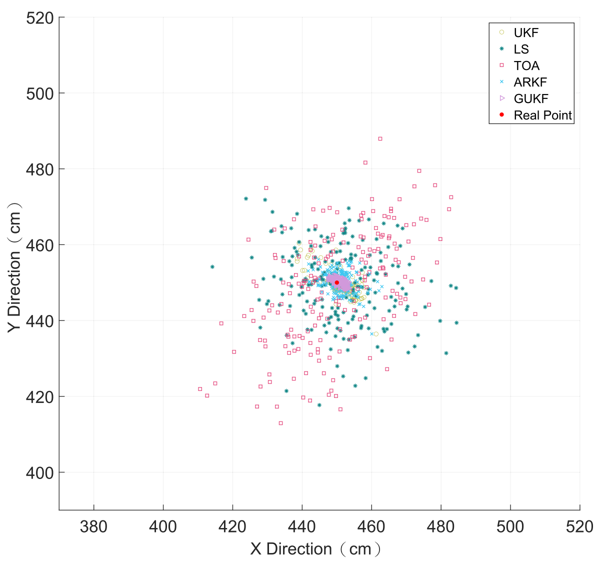

FIGURE 3 Static simulation trajectory of tags.

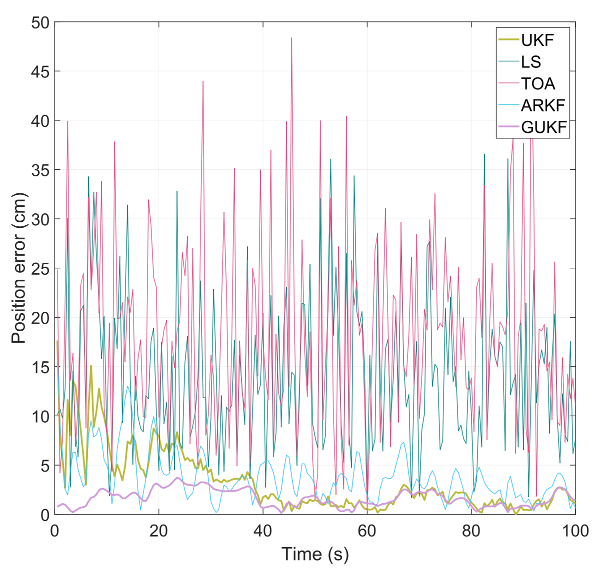

FIGURE 4 Static simulation error of tags.

TABLE 1 Mean and root-mean-square errors of static simulation experiments.

Method	Mean (cm)	RMSE (cm)
UKF	3.2797	4.5574
LS	14.0573	16.0183
TOA	18.4620	20.7913
ARKF	3.8268	6.4628
GUKF	1.6303	1.8438

Abbreviations: ARKF, adaptive robust Kalman algorithm; LS, Least Squares; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

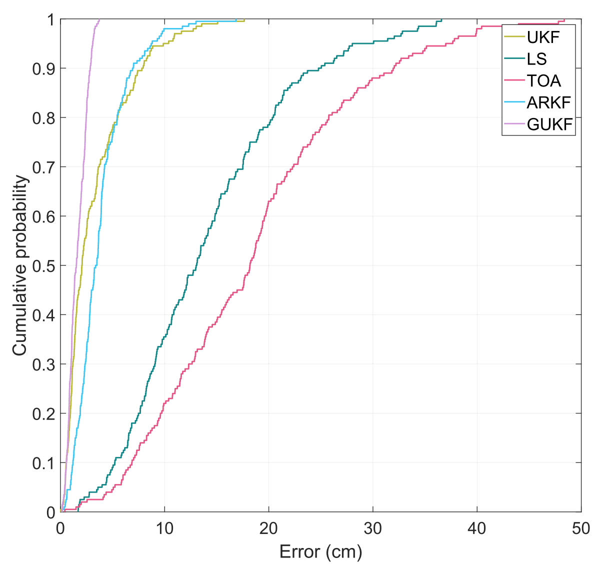

FIGURE 5 Cumulative distribution function curve of static simulation experiment.

static simulation experiment, it can be concluded that the GUKF algorithm has a high precision of positioning solution.

Figure 6 is the simulated trajectory of the UWB tag in linear motion, and the red point in the figure is its initial point. It can be found that the predicted trajectories of LS and TOA are obviously too dispersed, while the trajectory of ARKF is slightly better, but there are some excessive offsets, and UKF shrinks after some fluctuations at the initial point, and the trajectory of GUKF is the closest to the real trajectory compared to the rest of the algorithms.

From the error curve shown in Figure 7, the fluctuation range of the error curve of GUKF algorithm and UKF algorithm is relatively small, but the error margin value of GUKF algorithm proposed in this paper is lower than that of UKF algorithm for a period of time, and the other periods are approximately coincident.

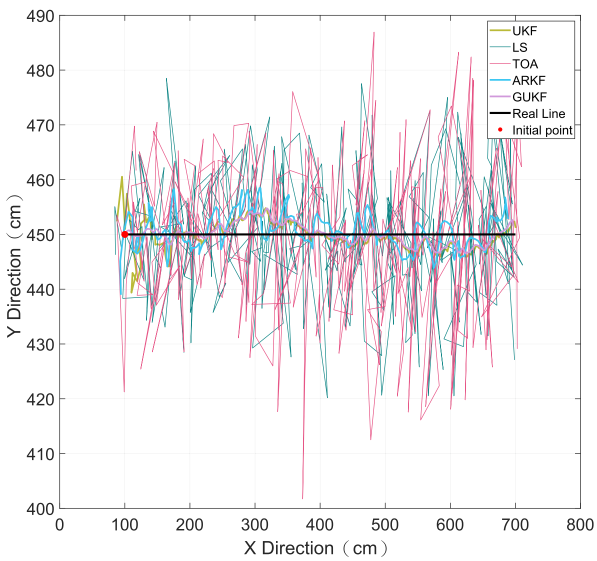

FIGURE 6 Dynamic simulation trajectory in linear motion of the tag.

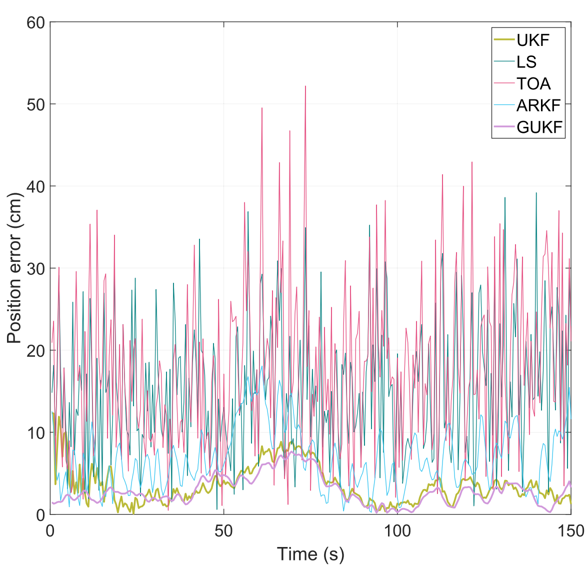

FIGURE 7 Dynamic simulation in linear motion error of tags.

Table 2 shows the mean error and RMSE of different localisation algorithms in the dynamic simulation experiments. From the table, it can be seen that the GUKF algorithm performs the best in terms of mean error and RMSE, with a mean error of 2.8911 cm and a RMSE of 3.3914 cm, which is significantly lower than the other algorithms. The UKF algorithm has a mean error of 3.5880 cm and a RMSE of 4.2794 cm, which is also good, but there is a significant gap compared with GUKF. The LS algorithm has a RMSE of 15.3680 cm and a RMSE of 17.2710 cm, indicating that it performs poorly in dealing with dynamic non-linear problems. The TOA algorithm has a RMSE of 17.8274 cm and a RMSE of 20.2679 cm, which is the highest among all the algorithms, suggesting that it has low localisation accuracy in dynamic environments. The ARKF algorithm has a RMSE of 6.8504 cm and a RMSE of 4.2794 cm. which is an improvement in terms of robustness, but there is still a big gap compared with GUKF.

Figure 8 is the CDF of the algorithm obtained under simulated linear motion, and it can be intuitively found that

TABLE 2 Mean and root-mean-square errors of dynamic simulation in linear motion experiments.

Method	Mean (cm)	RMSE (cm)
UKF	3.5880	4.2794
LS	15.3680	17.2710
TOA	17.8274	20.2679
ARKF	6.8504	8.8841
GUKF	2.8911	3.3914

Abbreviations: ARKF, adaptive robust Kalman algorithm; LS, Least Squares; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

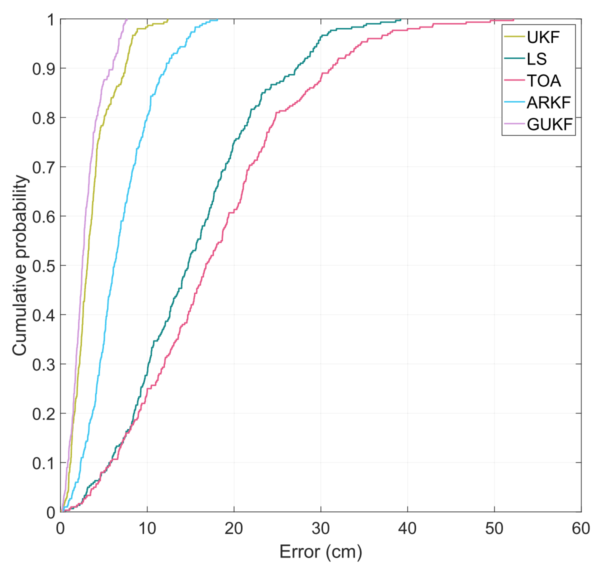

FIGURE 8 Cumulative distribution function curve of simulation in linear motion.

UKF and GUKF perform well in this experiment, and the maximum error of GUKF is not more than 10 cm.

Figure 9 is the simulated trajectory of the UWB tag in square motion, and the red point in the figure is its initial point. From the figure, it can be seen that several algorithms have certain error oscillations under the square motion, which are overall in the vicinity of the real trajectory, and it is not possible to visually compare the advantages and disadvantages of several algorithms.

From Figure 10 it is observed that initially the UKF and GUKF errors are large and then stabilise below 20 cm, while LS and TOA continue to have large variations in error, as for the ARKF algorithm it is more stable, with the error varying around 10 cm.

Table 3 shows the mean error and RMSE of different positioning algorithms in the square motion simulation. From the table, it can be seen that GUKF algorithm has the best performance in terms of RMSE 9.0972 cm and RMSE 10.5911 cm, which is significantly better than the other algorithms. UKF algorithm has the RMSE 11.9051 cm and RMSE 13.8322 cm; although it also performs well, but it is still not as good as the GUKF. LS algorithm has the RMSE 17.1670 cm and RMSE 19.7961 cm, which shows that it performs poorly in dynamic non-linear problems. The LS algorithm has a mean error of 17.1670 cm and RMSE of 19.7961 cm, which indicates that it performs poorly in dynamic non-linear problems. The TOA algorithm has a mean error of 16.7490 cm and a RMSE of 19.0892 cm, which is higher than all the other algorithms, indicating that it is not as accurate as the GUKF algorithms in dynamic environments. The ARKF algorithm has a mean error of 10.6589 cm and RMSE of 17.9840 cm, although it has improved robustness. Although the robustness of the ARKF algorithm has been improved, it still has a certain gap compared with the GUKF algorithm.

Figure 11 shows the error performance of different localisation algorithms under the CDF. From the figure, it can be seen that the GUKF algorithm has the highest accuracy until 0.9 probability, after which the ARKF algorithm is better

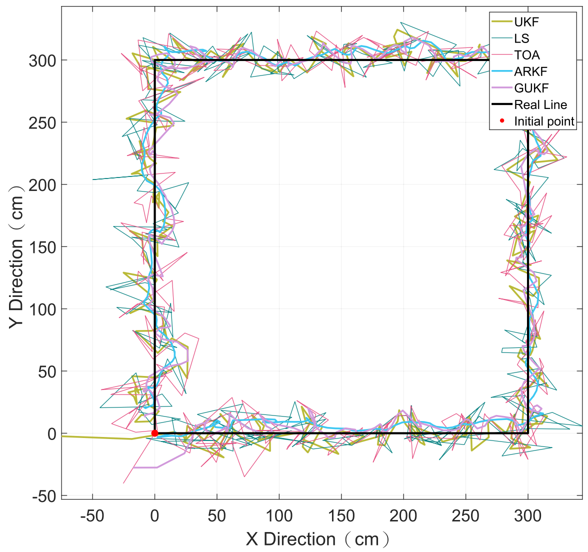

FIGURE 9 Dynamic simulation trajectory in square motion of the tag.

TABLE 3 Mean and root-mean-square errors of dynamic simulation in square motion experiments.

Method	Mean (cm)	RMSE (cm)
UKF	11.9051	13.8322
LS	17.1670	19.7961
TOA	16.7490	19.0892
ARKF	10.6589	17.9840
GUKF	9.0972	10.5911

Abbreviations: ARKF, adaptive robust Kalman algorithm; LS, Least Squares; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

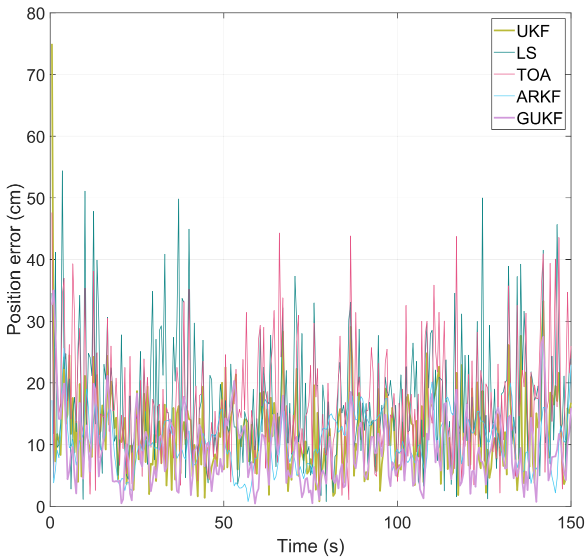

FIGURE 10 Dynamic simulation in square motion error of tags.

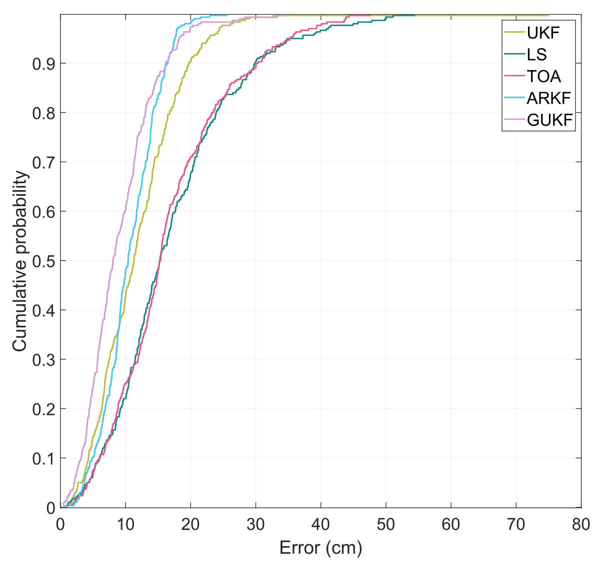

FIGURE 11 Cumulative distribution function curve of simulation in square motion.

than the GUKF algorithm. In contrast, UKF is not as good as previous experiments, while LS and TOA remain less effective.

Figure 12 is the simulated trajectory of the UWB tag in ellipse motion, and the red point in the figure is its initial point. From the figure, it can be seen that the two algorithms ARKF and GUKF are better than the rest of the algorithms under ellipse.

In the error map represented by Figure 13, LS, TOA and GUKF oscillations are too large and frequent, and the GUKF and ARKF algorithms also have frequent oscillations but smaller errors than the rest of the algorithms.

Table 4 shows the mean error and RMSE of different localisation algorithms in the elliptical motion simulation experiments. It can be seen that the GUKF algorithm performs the best in terms of RMSE 7.6776 cm and RMSE 8.9392 cm, which is significantly better than the other algorithms. The UKF algorithm has a RMSE of 11.1142 cm and RMSE 12.5398 cm, which is not as good as the GUKF algorithm. The LS algorithm has a RMSE of 14.1904 cm and a RMSE of 15.9874 cm, which indicates that it is not as good as the GUKF algorithm in the dynamic simulation experiment, showing that its performance is poor when dealing with dynamic non-linear problems. The TOA algorithm has a RMSE of 18.1984 cm and a RMSE of 21.0724 cm, which is higher than all the other algorithms, which shows that its localisation accuracy is low in dynamic environments. The ARKF algorithm has a RMSE of 8.0489 cm and a RMSE of 15.0724 cm, which shows that its localisation accuracy is low when dealing with dynamic environments. RMSE is 15.3641 cm, which is still a gap compared with GUKF despite the improvement in robustness.

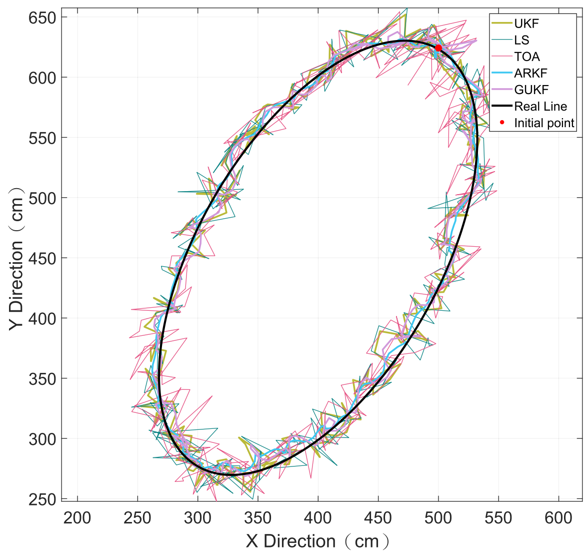

FIGURE 12 Dynamic simulation trajectory in ellipse motion of the tag.

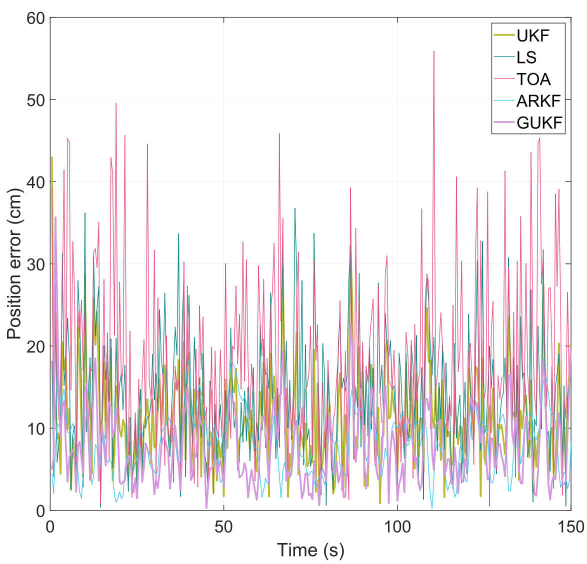

FIGURE 13 Dynamic simulation in ellipse motion error of tags.

As can be seen from Figure 14, the GUKF method exhibits lower positioning errors in most cases, which suggests that the algorithm is able to provide more accurate positioning results. In contrast, the UKF, LS and TOA algorithms have a wider range of errors, especially in some specific conditions, where the maximum error is close to 50 cm. In addition, the ARKF algorithm provides relatively good performance even

TABLE 4 Mean and root-mean-square errors of dynamic simulation in ellipse motion experiments.

Method	Mean (cm)	RMSE (cm)
UKF	11.1142	12.5398
LS	14.1904	15.9874
TOA	18.1984	21.0724
ARKF	8.0489	15.3641
GUKF	7.6776	8.9392

Abbreviations: ARKF, adaptive robust Kalman algorithm; LS, Least Squares; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

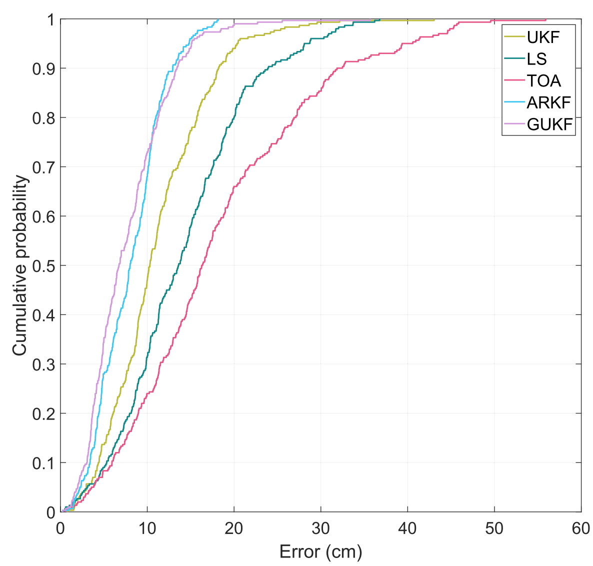

FIGURE 14 Cumulative distribution function curve of simulation in ellipse motion.

though its error is higher than that of the GUKF in some conditions.

By comparing different localisation algorithms in static and dynamic simulation experiments, it can be concluded that the GUKF algorithm exhibits the highest localisation accuracy and stability in all experiments. The GUKF algorithm has the lowest mean error and RMSE in both static simulation and dynamic simulation for linear, square and elliptical motions. In the CDF plot, the CDF curve of GUKF is always at the top, indicating that it has the highest probability of having smaller errors in all error intervals. In contrast, the UKF algorithm performs well but is slightly inferior to the GUKF, the ARKF algorithm performs better in some error intervals but is overall inferior to the GUKF, and the TOA and LS algorithms have lower localisation accuracies in complex environments. In summary, the GUKF algorithm shows the highest positioning accuracy and stability in all the experiments and is suitable for application scenarios that require high-precision positioning.

4.2 | Field experiment

In order to verify the effect of the GUKF algorithm proposed in this paper in practical application, the hardware equipment of UWB positioning system is developed by ourselves. The actual positioning environment is composed of 4 base stations. As shown in Figure 15, the actual experimental environment and self-made positioning hardware equipment are used. Base station BS1 is set as the primary base station, the other three base stations are set as the secondary base stations, and the bit tag is set as Tag. Static and dynamic experimental data are collected in the positioning area. In the actual experiment, the experimental data were collected in LOS and NLOS environment, respectively.

4.2.1 | Measured static experiments in LOS and NLOS environments

Before the start of the experiment, coordinate calibration of four base stations was carried out. The coordinate of base station 1 was set as the origin of the positioning region, and the other three base stations were calibrated according to a certain distance using a laser rangefinder with base station 1 as the reference point. The ranging error of laser rangefinder was $\pm 1cm$. The actual coordinates of the four base stations are as follows: $BS1(0,0)$, $BS2(480,0)$, $BS3(480,640)$, $BS4(0,560)$, the unit is cm. In the static experiment, the positioning tag is randomly placed on a measuring point within the positioning area. After accurate measurement, it can be seen that the real coordinate of the measuring point is (240, 323), and the unit is cm. The real measuring point is mainly used to measure the positioning error. In the experiment, static experiments were carried out in LOS and NLOS hardware device environments, respectively. After collecting the original data information for a period of time, GUKF, UKF, LS, ARKF and TOA algorithms were used to solve the problems, respectively. Figure 16 shows the location tag solution path in LOS environment. Figure 17 shows the location tag solution path in NLOS environment. It can be seen from the two trajectory diagrams that in the static experimental test of the two environments, the location estimation calculated by GUKF algorithm is more dense and closer to the true value than the other four location algorithms. Therefore, from the positioning trajectory of the five algorithms, the solution accuracy of GUKF algorithm is higher,

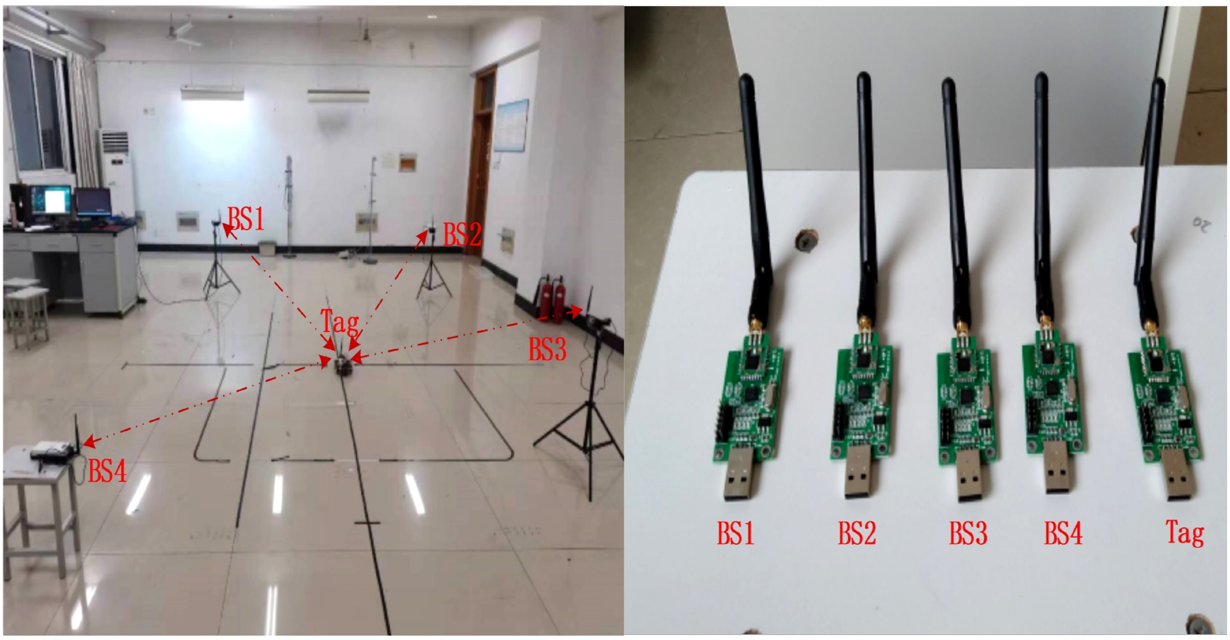

FIGURE 15 Experimental environment and homemade positioning hardware equipment.

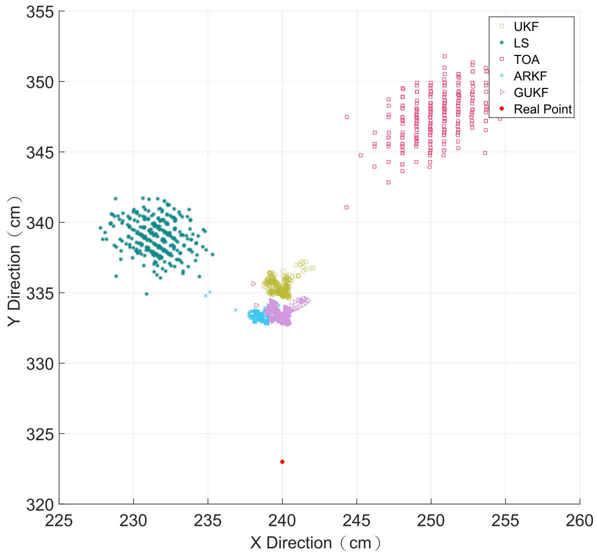

FIGURE 16 The location tag solution path in Line-of-Sight environment.

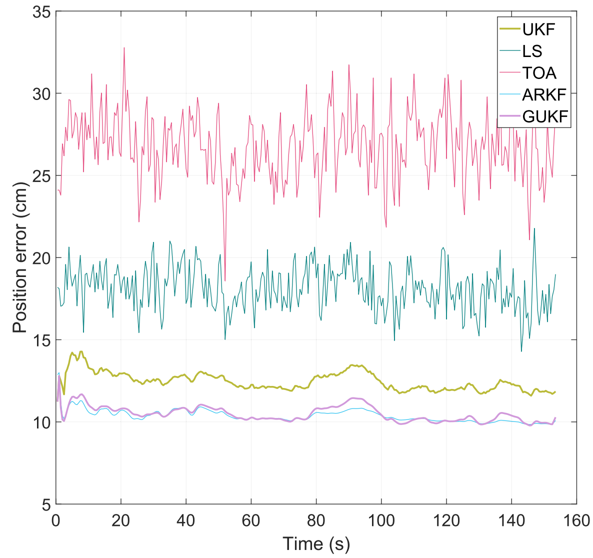

FIGURE 18 Static experimental errors of different algorithms in Line-of-Sight environments.

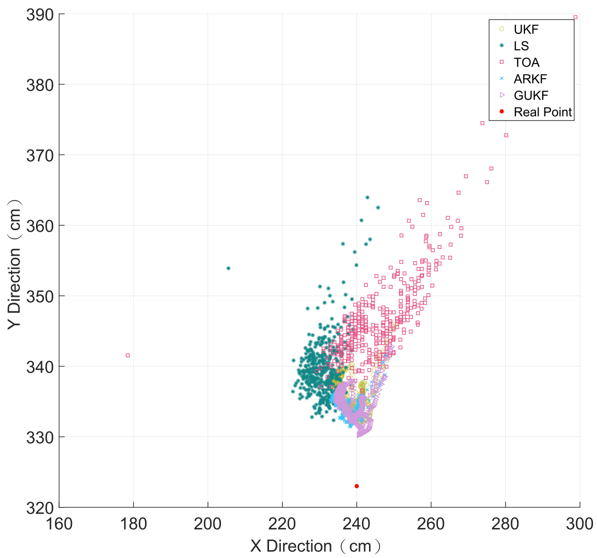

FIGURE 17 The location tag solution path in Non-Line-of-Sight environment.

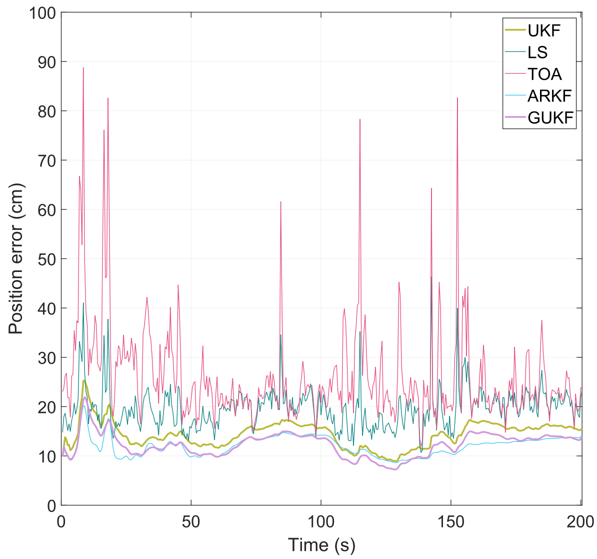

FIGURE 19 Static experimental errors of different algorithms in Non-Line-of-Sight environments.

and the positioning performance is better than that of other algorithms.

Figures 18 and 19 shows the static experimental error curves of five different algorithms in LOS and NLOS environments, respectively. The Euclid distance between the coordinate points calculated by five different algorithms and the real coordinate points (240, 323) at different times is calculated, and the error curves shown in Figures 18 and 19 are drawn in chronological order. It can be seen from the error curves of the two static experiments that the error curve of GUKF algorithm is below the error curve of the other four positioning algorithms. It can be seen that the positioning accuracy of GUKF algorithm is higher than that of the other four algorithms.

Table 5 shows the mean error and RMSE of UKF, LS, TOA, ARKF and GUKF in LOS and NLOS environments. From the table, it can be seen that the GUKF algorithm shows optimal performance in both environments. Specifically, in the LOS environment, the GUKF method has an average error of 7.6776 cm and an RMSE of 8.9392 cm, which are significantly lower than the other methods. While in the NLOS environment, the GUKF method also maintains the lowest average error of 7.6776 cm and RMSE of 8.9392 cm, showing its robustness in complex environments. In contrast, the TOA method performs the worst in both environments, with mean errors and RMSEs of 18.1984 and 21.0724 cm (LOS), and 18.1984 and 21.0724 cm (NLOS environment), respectively. This indicates that the TOA method has significant limitations in dealing with multipath effects and non-line-of-sight (NLOS) propagation. The UKF and ARKF algorithms perform more closely in both environments, but are slightly inferior to the GUKF method in general. In the LOS environment, the UKF algorithm has an average error of 11.1142 cm and an RMSE of 12.5398 cm, while the ARKF algorithm has an average error of 8.0489 cm and an RMSE of 15.3641 cm. In the NLOS environment, both algorithms perform in line with the results in the LOS environment.

Figures 20 and 21 are CDF plots for LOS and NLOS environments, respectively. The largest error in the LOS environment is the TOA algorithm, which reaches a maximum of near 35 cm, while the CDF curves of the two algorithms, ARKF and GUKF, are highly coincident and the errors are all near 10 cm; in the NLOS environment, the ARKF and GUKF algorithms are between 10 and 20 cm, with a slight decrease in the accuracy compared to the LOS environment, and the TOA algorithm has a maximum error of near 90 cm.

Based on the actual static experimental positioning effect of positioning tags in LOS and NLOS environments, the positioning trajectory, error curve, RMSE and CDF curve of the five positioning algorithms can be evaluated to show that GUKF algorithm has better positioning performance, and also has better positioning effect in NLOS experimental environment.

TABLE 5 Mean error and RMSE of static experiments in LOS/NLOS environments.

Environment	Method	Mean (cm)	RMSE (cm)
LOS	UKF	11.1142	12.5398
	LS	14.1904	15.9874
	TOA	18.1984	21.0724
	ARKF	8.0489	15.3641
	GUKF	7.6776	8.9392
NLOS	UKF	11.1142	12.5398
	LS	14.1904	15.9874
	TOA	18.1984	21.0724
	ARKF	8.0489	15.3641
	GUKF	7.6776	8.9392

Abbreviations: ARKF, adaptive robust Kalman algorithm; LOS, Line-of-Sight; LS, Least Squares; NLOS, Non-Line-of-Sight; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

4.2.2 | Observed dynamic experiments in LOS and NLOS environments

Experiments were carried out on the positioning tags in the self-built real experimental environment, which was divided into LOS and NLOS environments, and the coordinates of the four base stations were the same as those of the static experiment. Before the experiment starts, a standard trajectory is selected as the true value in the positioning region to facilitate error analysis with the estimation of the positioning algorithm.

The black bold line in Figures 22 and 23 indicates the real track in the location area, with the start point (120, 320) and the end point (360, 320) of the track. At the beginning of the experiment, the positioning tag was installed on the mobile car from the start point to the end point at a constant speed, and the real track was evenly divided according to the number of measured data points, and the equal-point was used as the real point and the calculated coordinate value for error calculation.

Figure 22 shows the dynamic trajectory diagram of the positioning tag in LOS environment, and Figure 23 shows the dynamic trajectory diagram of the positioning tag in NLOS environment. From the dynamic estimation diagram of the two experimental environments, it can be seen that the positioning

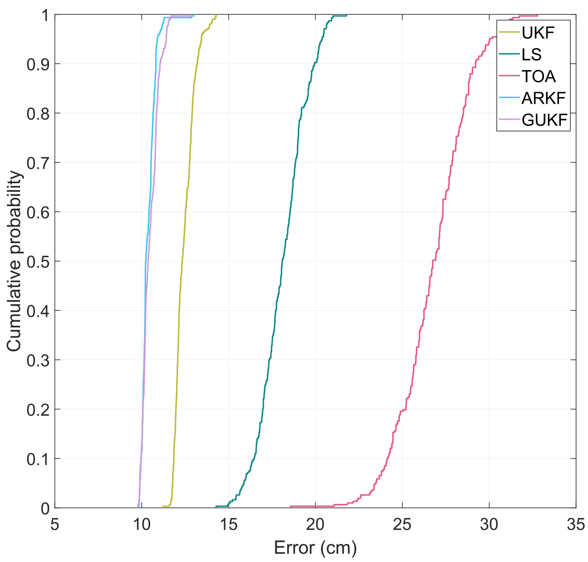

FIGURE 20 Static experimental errors of different algorithms in Line-of-Sight environments.

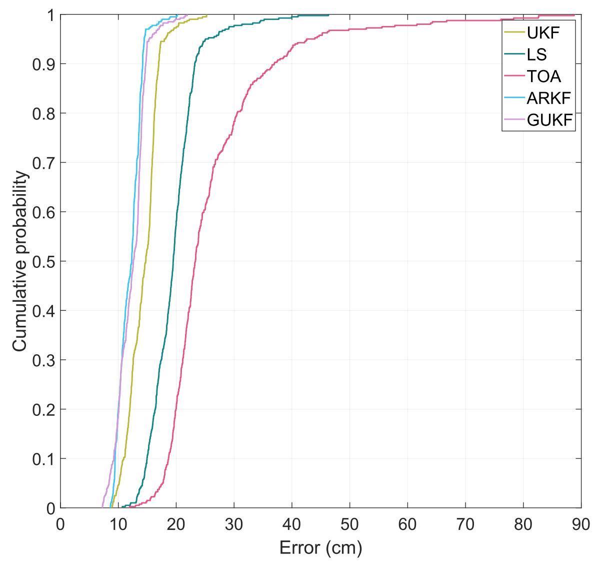

FIGURE 21 Static experimental errors of different algorithms in Non-Line-of-Sight environments.

estimation of the GUKF algorithm and the UKF algorithm among the five positioning algorithms is smooth and close to the true value.

The calculated values of the five positioning algorithms and the real values in the two experimental environments were used to calculate the error and draw the error curve. Figure 24 shows the error curve of the positioning tag in the LOS environment. The errors of LS, TOA and ARKF algorithms basically remain between 10 and 20 cm, with occasional errors close to 30 cm. The trajectories of the UKF and GUKF algorithms are basically similar and have the smallest error, but in the vast majority of cases GUKF is better than UKF.

Figure 25 shows the error curve of the positioning tag in the NLOS environment. It can be seen from the figure that the positioning accuracy of LS and TOA algorithms is greatly affected by the NLOS environment, and the amplitude value of the error curve fluctuates greatly. Although GUKF algorithm and UKF algorithm are affected by NLOS environment, their own advantages can play a better filtering role. From the error curves of these two filtering algorithms, it can be seen that GUKF algorithm has a slight advantage over UKF algorithm. The specific advantages will be verified according to RMSE and CDF curves.

Table 6 shows the average error and RMSE of the five localisation algorithms for dynamic experiments in LOS and NLOS environments. As can be seen from the table, the GUKF algorithm shows optimal performance in both environments. In the LOS environment, the GUKF algorithm

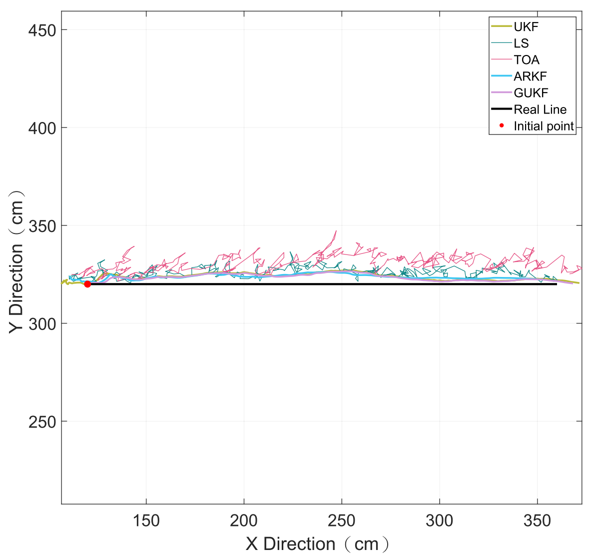

FIGURE 22 Dynamic experiment trajectories of different algorithms in Line-of-Sight environments.

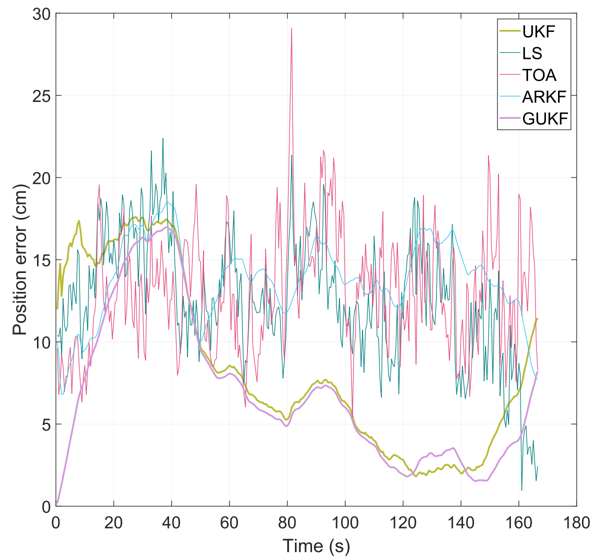

FIGURE 24 Dynamic experimental errors of different algorithms in Line-of-Sight environments.

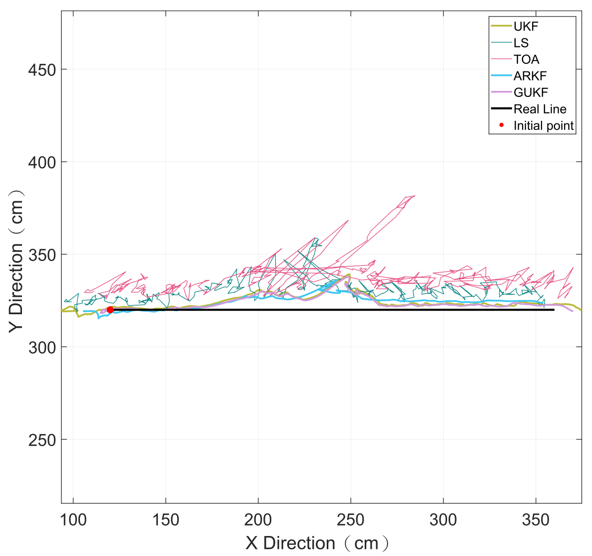

FIGURE 23 Dynamic experiment trajectories of different algorithms in Non-Line-of-Sight environments.

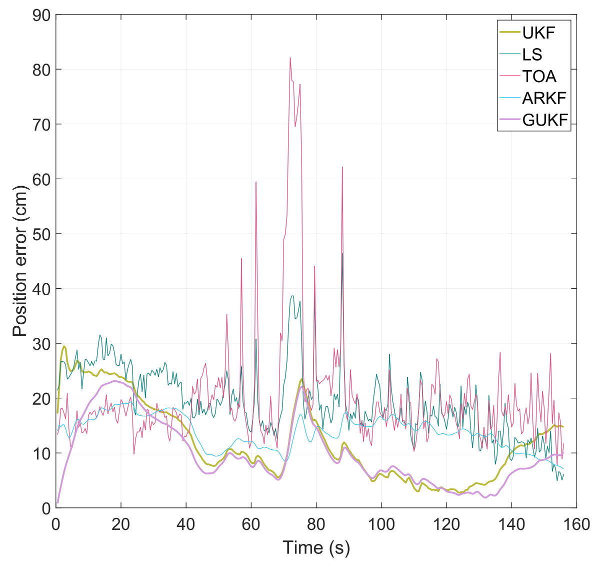

FIGURE 25 Dynamic experimental errors of different algorithms in Non-Line-of-Sight environments.

TABLE 6 Mean error and RMSE of dynamic experiments in LOS/NLOS environments.

Environment	Method	Mean (cm)	RMSE (cm)
LOS	UKF	8.4636	9.9633
	LS	12.6203	13.1787
	TOA	13.0368	13.5219
	ARKF	13.7650	17.1679
	GUKF	6.9379	8.2989
NLOS	UKF	12.1576	14.0939
	LS	19.3971	20.4466
	TOA	20.2719	23.2535
	ARKF	13.8012	19.9519
	GUKF	9.9483	11.5715

Abbreviations: ARKF, adaptive robust Kalman algorithm; LOS, Line-of-Sight; LS, Least Squares; NLOS, Non-Line-of-Sight; RMSE, Root Mean Square Error; TOA, Time of Arrival; UKF, Unscented Kalman Filter.

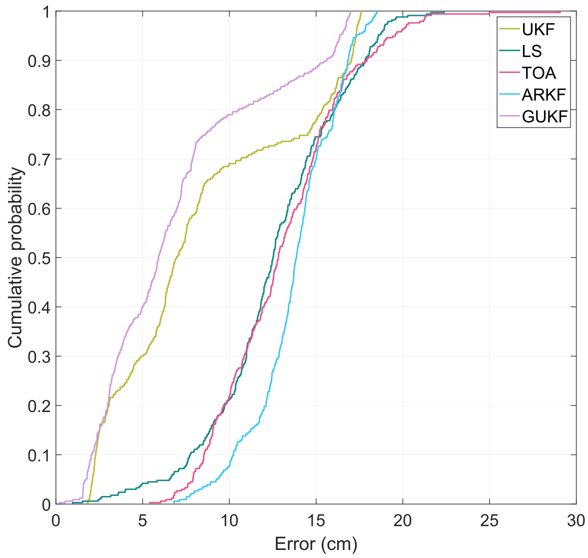

FIGURE 26 Cumulative distribution function curves of dynamic experimental errors of different algorithms in Line-of-Sight environments.

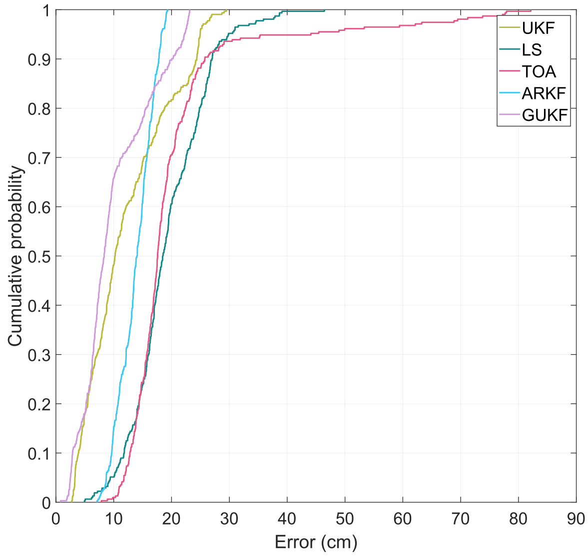

FIGURE 27 Cumulative distribution function curves of dynamic experimental errors of different algorithms in Non-Line-of-Sight environments.

has an average error of 6.9379 cm and an RMSE of 8.2989 cm, which are significantly lower than the other algorithms. In contrast, the ARKF algorithm has the highest mean error and RMSE of 13.7650 and 17.1679 cm, respectively. The UKF algorithm has the next best performance with a mean error of 8.4636 cm and an RMSE of 9.9633 cm. The LS and TOA algorithms perform in the middle of the range but are generally inferior to the GUKF and UKF algorithms. In the NLOS environment, the GUKF algorithm still exhibits the lowest mean error and RMSE of 9.9483 and 11.5715 cm, respectively. In contrast, the TOA algorithm has the highest mean error and RMSE of 20.2719 and 23.2535 cm, respectively. The LS algorithm also performs poorly, with a mean error of 19.3971 cm and an RMSE of 20.4466 cm. The UKF and ARKF algorithms perform in the middle, but are generally inferior to the GUKF algorithm.

The CDF curves of five localisation algorithms are drawn by the error values of the localisation tags in LOS and NLOS environments at different times to measure the localisation performance of the five localisation algorithms. Figure 26 shows the CDF curve of the experimental data in LOS environment solved by five positioning algorithms, and Figure 27 shows the CDF curve of the experimental data in NLOS environment solved by five positioning algorithms. GUKF's CDF in the LOS environment is essentially in the upper part of the rest of the algorithms, indicating that it is used for better accuracy. ARKF, on the other hand, has the worst results in terms of error under this experiment instead, but its maximum error is less than 20 cm, which is still better than the LS and TOA algorithms. In the NOLS environment, the maximum error of all four algorithms except ARKF increased, the worst being the TOA algorithm with an error of 80 cm. GUKF, although its error increased, also had a percentage of 0.8 lower than the error of ARKF.

5 | CONCLUSION

In this paper, to solve the problem of non-linear model positioning accuracy in UWB indoor positioning system, we propose a Gaussian untraceable KF algorithm based on Gaussian smoothing filter. The algorithm improves the positioning accuracy of the UWB indoor positioning system by using a data processing weighted template designed with Gaussian functions in combination with the UKF. In both static and dynamic tests, the GUKF algorithm achieves an RMSE reduction rate of 15.88% and 16.67% in LOS and NLOS environments, respectively, compared to the conventional UKF algorithm. Furthermore, the localisation performance of the GUKF algorithm under LOS and NLOS conditions is significantly better than existing algorithms such as LS, TOA and ARKF. We verify the effectiveness of the GUKF algorithm through simulation experiments and real measurement experiments. The experimental results show that the GUKF algorithm provides more accurate position estimation under LOS and NLOS conditions, which is confirmed by a number of metrics such as mean error, CDF curve and RMSE. The research in this thesis not only improves the performance of UWB indoor positioning algorithms, but also provides valuable references for research in other related fields.

The practical significance of this study lies in the fact that the GUKF algorithm can significantly improve the accuracy of indoor positioning systems, which is of significant application value for application scenarios requiring high-precision positioning (e.g., smart logistics, security surveillance, medical care, etc.). Based on the results of this study, future research directions could include exploring the possibility of combining the GUKF algorithm with other state-of-the-art filtering techniques, as well as experimenting with the combination of positioning algorithms and machine learning algorithms in more complex and dynamically changing environments.

At the same time, we recognise that there are some limitations to this research study. First, although the GUKF algorithm showed good performance in the experiments, its computational complexity is high, which may affect its application on devices with limited computational resources. Secondly, the experiments were mainly conducted under specific environmental conditions, and the adaptability to a wider range of environments needs further verification. Therefore, one of the future directions for improvement is to optimise the GUKF algorithm to reduce its computational cost while maintaining or improving its positioning accuracy, so that it can better serve practical application scenarios.

AUTHOR CONTRIBUTIONS

DaLong Sun: Conceptualization; data curation; formal analysis; investigation; methodology; software; validation; visualization; writing—original draft; writing—review and editing. **Mingsheng Wei**: Supervision; writing—review and editing. **Yiyang Lyu**: Conceptualization; investigation; methodology; software; validation; visualization; writing—original draft. **Di Wang**: Supervision; writing—review and editing. **Shidang Li**: Formal analysis; validation; writing—review and editing. **Wenshuai Li**: Conceptualization; methodology; visualization; supervision. **Lei He**: Formal Analysis; validation; writing—review and editing. **Shihu Zhu**: Supervision; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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